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CMPU 250

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Project Proposal

Research Questions:

How accurately do MLB Pipeline prospect rankings predict future MLB performance?

Does the predictive accuracy of MLB Pipeline prospect rankings differ between pitchers and position players, and for position players, does it differ by position?

Are certain player positions/characteristics systematically over or underestimated by MLB pipeline rankings relative to their actual MLB performance?

Introduction:

Evaluating young talent is a pivotal, but also incredibly difficult part of building a successful Major League Baseball (MLB) organization. Teams are required to make decisions about players with little certainty about how those skills will translate to the MLB level. This makes scouting and player projection very uncertain and largely predictive. Almost all MLB players spend time developing in Minor League Baseball before reaching the major leagues, moving up through different levels as they improve. Previous research has examined how well information about young players can predict future MLB success. Conforti et al. (2022) analyzed 1,623 MLB players drafted between 2005 and 2015 and found that players selected earlier in the MLB draft accumulated more Wins Above Replacement (WAR, a stat that compares players to a replacement-level player). This suggested that evaluations made before a player even starts their professional career contain meaningful information about their future performance. Chandler and

Stevens (2012) examined whether minor-league statistics could predict future MLB success, using statistical techniques to identify which measures were most useful for evaluating developing players. This demonstrates both the potential and difficulty that comes with using information from earlier stages of a player's career to predict future MLB performance. MLB Pipeline, [MLB.com](https://mlb.com)'s prospect analysis service, provides rankings of the top 100 prospects in professional baseball each year. They rank minor league and budding major league players on scouting evaluations, statistics, and projections of future potential. Building on previous research, our project focuses specifically on how MLB Pipeline prospect rankings predict subsequent MLB performance and whether their predictive accuracy differs systematically across different types of players or different characteristics.

Motivation:

Jack, Dash, and Miller are all baseball players who closely watch their favorite teams. While signing top free agents and trading for star players can be very advantageous, a major factor separating successful and unsuccessful organizations is how well they identify young talent and develop them into MLB-caliber contributors. Evaluating prospects is extremely difficult, however, as players vary in age, position, level of competition, injury history, and many other characteristics. MLB pipeline tries to evaluate these players by ranking the top prospects in baseball each year, but many highly ranked prospects fail to have a successful career in the MLB, and many lower-ranked prospects greatly outperform their evaluations and become impact players in the MLB. This made us wonder how accurately these rankings predict future MLB performance, and if accuracy differs by certain characteristics or types of players. By examining where these predictions are strongest and weakest, we hope to gain a greater understanding of the strengths and weaknesses of prospect evaluation in baseball. While previous research examined

how factors like draft position and minor-league performance predict MLB success, less research has been conducted on whether publicly available rankings accurately predict MLB performance across various types of players.

Data:

We plan to create a dataset using MLB Pipeline's annual Top 100 prospect rankings from 2015 through 2020. These rankings will provide approximately 600 player-year observations (100 players x 6 years), although some players may appear in the Top 100 in multiple years. We will collect each player's name, ranking year, prospect rank (1-100), position, and MLB organization. Prospect rank is a discrete numerical variable, and position and organization are categorical variables. We will combine these rankings with MLB performance data from Baseball Savant. Performance variables may include whether the player reached MLB, games played, and certain measures of performance such as WAR (Wins Above Replacement), as well as more detailed stats specific to position players or pitchers such as plate appearances, hits, home runs, and batting average for position players, and innings, ERA, and appearances for pitchers, allowing us to stay comfortably above 10 variables. Since they are evaluated using different statistics, we will use WAR as our primary data point when comparing position players to pitchers.

The MLB Pipeline rankings will be collected from historical [MLB.com](https://mlb.com) Top 100 lists, while MLB performance data will be obtained from Baseball Savant. We will clean players' names and positions so that players can be matched across the datasets, and will account for players who appear in multiple years of the prospect rankings. We are also considering using a fixed period of MLB performance following each ranking so that players ranked in earlier years don't have an unfair advantage by having more time to accumulate statistics.

Methods:

We will start with exploratory data analysis to evaluate the distribution of prospect rankings and subsequent MLB performance. We will use visualizations like scatterplots to examine the relationship between prospect rank and MLB performance, and box plots to compare outcomes across positions and player types. In order to determine how well MLB pipeline rankings predict subsequent MLB performance, we will use regression analysis to examine the relationship between ranking and later MLB success. We may also use classification methods to examine whether prospect rank predicts whether a player ultimately reaches the MLB. To determine whether ranking accuracy differs between pitchers and position players, we will separate players into these groups and compare the relationship between prospect ranking and MLB performance for each group. Finally, we will compare ranking errors across individual positions to determine whether certain positions/standout traits tend to perform better or worse than their prospect rankings would suggest.

Research Expectations:

We expect higher-ranked MLB Pipeline prospects to perform better in the MLB than lower-ranked ones, but we also expect significant variation in outcomes. We also expect accuracy of these rankings to differ between pitchers and position players, and across individual positions as well. If certain groups/traits are consistently evaluated better or worse than the rankings would predict, it could suggest that MLB Pipeline is more accurate at certain types of players than others, and if there are no significant differences, it would suggest that MLB Pipeline's rankings are consistent across different types of players.

Bibliography:

Chandler, Gabriel, and Guy Stevens. "An Exploratory Study of Minor League Baseball Statistics." *Journal of Quantitative Analysis in Sports*, vol. 8, no. 4, 2012, pp. 1–28.

<https://www.degruyterbrill.com/document/doi/10.1515/1559-0410.1445/html>

Conforti, Christian M., Ryan L. Crotin, and Jordan Oseguera. "Major League Draft WARs: An Analysis of Wins Above Replacement in Player Selection." *Journal of Sports Analytics*, vol. 8, no. 1, 2022, pp. 77–84. <https://journals.sagepub.com/doi/10.3233/JSA-200586>

Work Agreement:

We've all agreed to contribute equally in terms of quantity and quality. For bigger sections and decisions, we will all work together. We also have a groupchat, and for smaller work, we can do work individually, and check with both other group members for approval. We will all do the code and cleaning together in order to make sure each group member understands the methodology and processes used. We also plan to complete all work ahead of deadlines in order to give the other members time to verify and approve of the work. Before edits and changes are made, we will notify other group members and get approval before putting them into place.